WORKED SOLUTION

STEP I 2015, Question 5

Pure • Integrals with a Parameter

STEP I 2015, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/15-s1-q5

The key idea

Both integrals have elementary antiderivatives in t, with x held fixed as a parameter, so the instruction "by evaluating the integral" is the whole hint: find a closed form in x, then sketch that. The care lies in the details. In (i) the integrand can be unbounded at $t = 1$, and in (ii) the square roots evaluate to *absolute values*, which makes g a piecewise function with a surprisingly simple shape.

Solution

Part (i)

For fixed $x > 0$, an antiderivative of $(t-1)^{x-1}$ is $\frac{1}{x}(t-1)^x$. So

$$f(x) = \left[\frac{(t-1)^x}{x}\right]_1^3 = \frac{2^x - 0}{x} = \frac{2^x}{x}.$$

When $0 < x < 1$ the integrand $(t-1)^{x-1}$ blows up as $t \to 1^+$, so the integral is improper; but $(t-1)^x \to 0$ as $t \to 1^+$ because $x > 0$, so it converges and the value above stands.

Now sketch $y = 2^x/x$ for $x > 0$.

- As $x \to 0^+$, $2^x \to 1$, so $f(x) \to +\infty$: the y-axis is an asymptote.
- As $x \to \infty$, exponentials beat powers, so $f(x) \to \infty$ (and faster than any straight line).
- Differentiating, using $\frac{\mathrm{d}}{\mathrm{d}x}2^x = 2^x \ln 2$,

$$f'(x) = \frac{2^x \ln 2 \cdot x - 2^x}{x^2} = \frac{2^x(x \ln 2 - 1)}{x^2},$$

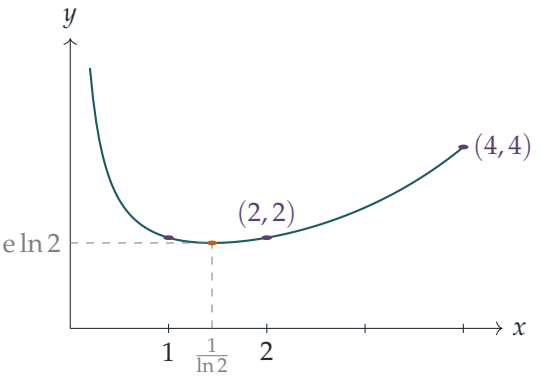
which is negative for $x < 1/\ln 2$ and positive for $x > 1/\ln 2$. So there is a single stationary point, a minimum, at $x = 1/\ln 2 \approx 1.44$.

- There, $2^{1/\ln 2} = \mathrm{e}^{(\ln 2)/\ln 2} = \mathrm{e}$, so the minimum value is

$$f\left(\frac{1}{\ln 2}\right) = \mathrm{e}\ln 2 \approx 1.88.$$

- Useful points: $f(1) = f(2) = 2$, $f(4) = 4$.

The curve is also convex: $\ln f(x) = x\ln 2 - \ln x$ is convex (its second derivative is $1/x^2 > 0$), and the exponential of a convex function is convex. So the sketch is a single U-shaped curve.



Part (ii)

First, the integrand makes sense: for $-1 \le t \le 1$,

$$1 - 2xt + x^2 \ge 1 - 2|x| + x^2 = (1 - |x|)^2 \ge 0,$$

with equality only when $|x| = 1$ and $t = \pm 1$ (an endpoint). We return to that case below.

For $x = 0$ the integrand is 1, so $g(0) = 2$. For $x \neq 0$, treat x as a constant: since $\frac{\mathrm{d}}{\mathrm{d}t}(1 - 2xt + x^2) = -2x$, an antiderivative is $-\frac{1}{x}\sqrt{1 - 2xt + x^2}$. Hence

$$g(x) = -\frac{1}{x}\Big[\sqrt{1 - 2xt + x^2}\Big]_{-1}^{1} = \frac{\sqrt{(1+x)^2} - \sqrt{(1-x)^2}}{x} = \frac{|1+x| - |1-x|}{x}.$$

? **Ask yourself** Why not simply write $\sqrt{(1+x)^2} = 1 + x$ and $\sqrt{(1-x)^2} = 1 - x$?

Because $\sqrt{u^2} = |u|$, not u. Doing it carelessly gives $g(x) = 2$ for every x, which is certainly wrong for large $|x|$, where the integrand is roughly $1/|x|$ and so $g(x) \approx 2/|x|$. We must split into cases.

- $-1 \le x \le 1$, $x \neq 0$: both $1 + x$ and $1 - x$ are ≥ 0, so $g(x) = \dfrac{(1+x) - (1-x)}{x} = 2$. This also agrees with $g(0) = 2$.

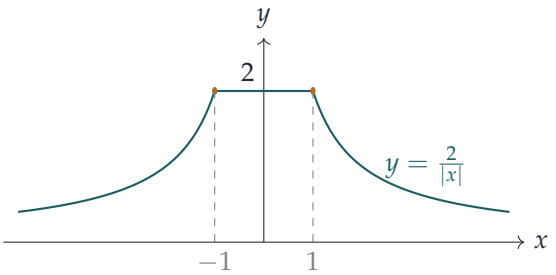
- $x > 1$: $|1+x| = 1+x$ and $|1-x| = x-1$, so $g(x) = \dfrac{2}{x}$.
- $x < -1$: $|1+x| = -1-x$ and $|1-x| = 1-x$, so $g(x) = \dfrac{-2}{x} = \dfrac{2}{|x|}$.

At $x = \pm 1$ the integrand is unbounded at one endpoint (for $x = 1$ it is $1/\sqrt{2-2t}$, which blows up as $t \to 1$), so the integral is improper. The antiderivative $-\sqrt{2-2t}$ is continuous up to $t = 1$, so it converges, and the formula still gives $g(\pm 1) = 2$.

So

$$g(x) = \begin{cases} 2, & |x| \le 1, \\ \dfrac{2}{|x|}, & |x| \ge 1. \end{cases}$$

The function is even (as it must be: replacing t by $-t$ shows $g(-x) = g(x)$). It is continuous everywhere, constant on $[-1, 1]$, and decays like $2/|x|$ towards the asymptote $y = 0$. At $x = \pm 1$ there is a corner: the one-sided gradients are 0 (inside) and ∓ 2 (outside), so g is not differentiable there.



Checks

- Part (i) at $x = 1$: the integrand is 1, so $f(1) = 2$, matching $2^1/1$. At $x = 2$ the integral is $\int_1^3 (t-1)\,\mathrm{d}t = 2$, matching $2^2/2$.
- Part (ii) at $x = 0$: the integrand is 1 and $g(0) = 2$. For large $|x|$ the integrand is close to $1/|x|$ over an interval of length 2, matching $2/|x|$.
- Numerically, $\int_1^3 (t-1)^{0.5}\,\mathrm{d}t = 1.8856\ldots = 2^{1.5}/1.5$; the integral in (ii) with $x = 1.2$ evaluates to $1.6667 = 2/1.2$, and with $x = 0.8$ to 2.0000.

Common slips

- Writing $\sqrt{(1-x)^2} = 1 - x$ for all x, and concluding that g is constant.
- Forgetting that the integral in (i) is improper for $0 < x < 1$, or that $(t-1)^x$ vanishes at $t = 1$ only because $x > 0$.
- Drawing g smooth at $x = \pm 1$: the two pieces meet with different gradients.

Beyond STEP

Part (ii) is Newton's shell theorem in disguise. With $t = \cos\theta$, the quantity $1/\sqrt{1-2xt+x^2}$ is the reciprocal distance from a point at distance x from the centre of a unit sphere to a point on the sphere, so g is (up to a constant) the gravitational potential of a uniform spherical shell: constant inside, and that of a point mass outside. The same expression is the generating function $\sum_n P_n(t)x^n$ of the Legendre polynomials, and $g(x) = 2$ for $|x| < 1$ reflects the fact that every P_n with $n \geq 1$ integrates to zero over $[-1, 1]$. Part (i), integrating t^{x-1} with x as a parameter, is the first step towards the Gamma and Beta functions. Mary Boas's *Mathematical Methods in the Physical Sciences* covers Legendre polynomials and the Gamma function at just the right level.

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